

```
In [1]: import pandas as pd
df = pd.read_csv(r'C:\Users\KHAN\Desktop\sp-500.csv')
print(df)
```

	Exchange	Symbol	Shortname	Longname \
0	NMS	AAPL	Apple Inc.	Apple Inc.
1	NMS	NVDA	NVIDIA Corporation	NVIDIA Corporation
2	NMS	MSFT	Microsoft Corporation	Microsoft Corporation
3	NMS	AMZN	Amazon.com, Inc.	Amazon.com, Inc.
4	NMS	GOOGL	Alphabet Inc.	Alphabet Inc.
..	...	...	...	...
497	NMS	CZR	Caesars Entertainment, Inc.	Caesars Entertainment, Inc.
498	NYQ	BWA	BorgWarner Inc.	BorgWarner Inc.
499	NMS	QRVO	Qorvo, Inc.	Qorvo, Inc.
500	NYQ	FMC	FMC Corporation	FMC Corporation
501	NYQ	AMTM	Amentum Holdings, Inc.	Amentum Holdings, Inc.

	Sector	Industry	Currentprice \
0	Technology	Consumer Electronics	254.49
1	Technology	Semiconductors	134.70
2	Technology	Software - Infrastructure	436.60
3	Consumer Cyclical	Internet Retail	224.92
4	Communication Services	Internet Content & Information	191.41
..	...	...	...
497	Consumer Cyclical	Resorts & Casinos	32.82
498	Consumer Cyclical	Auto Parts	31.88
499	Technology	Semiconductors	70.85
500	Basic Materials	Agricultural Inputs	50.15
501	Industrials	Specialty Business Services	19.17

	Marketcap	Ebitda	Revenuegrowth	City	State \
0	3846819807232	1.346610e+11	0.061	Cupertino	CA
1	3298803056640	6.118400e+10	1.224	Santa Clara	CA
2	3246068596736	1.365520e+11	0.160	Redmond	WA
3	2365033807872	1.115830e+11	0.110	Seattle	WA
4	2351625142272	1.234700e+11	0.151	Mountain View	CA
..	...	...	...	...	...
497	6973593600	3.668000e+09	-0.040	Reno	NV
498	6972155904	1.882000e+09	-0.048	Auburn Hills	MI
499	6697217024	6.731300e+08	-0.052	Greensboro	NC
500	6260525568	7.033000e+08	0.085	Philadelphia	PA
501	4664099328	4.330000e+08	-0.031	Chantilly	VA

	Country	Fulltimeemployees \
0	United States	164000.0
1	United States	29600.0

```

2    United States      228000.0
3    United States     1551000.0
4    United States      181269.0
..
497 United States      51000.0
498 United States      39900.0
499 United States       8700.0
500 United States       5800.0
501 United States       NaN

```

```

                                Longbusinesssummary  Weight
0    Apple Inc. designs, manufactures, and markets ... 0.069209
1    NVIDIA Corporation provides graphics and compu... 0.059350
2    Microsoft Corporation develops and supports so... 0.058401
3    Amazon.com, Inc. engages in the retail sale of... 0.042550
4    Alphabet Inc. offers various products and plat... 0.042309
..
497 Caesars Entertainment, Inc. operates as a gami... 0.000125
498 BorgWarner Inc., together with its subsidiarie... 0.000125
499 Qorvo, Inc. engages in development and commerc... 0.000120
500 FMC Corporation, an agricultural sciences comp... 0.000113
501 Amentum Holdings, Inc. provides engineering an... 0.000084

```

[502 rows x 16 columns]

```
In [2]: df = df[['Exchange', 'Sector', 'Industry', 'Currentprice', 'Marketcap', 'Ebitda', 'Revenuegrowth', 'Country', 'Fullt:
```

```
In [3]: print(df)
```

	Exchange	Sector	Industry \
0	NMS	Technology	Consumer Electronics
1	NMS	Technology	Semiconductors
2	NMS	Technology	Software - Infrastructure
3	NMS	Consumer Cyclical	Internet Retail
4	NMS	Communication Services	Internet Content & Information
..	...	...	...
497	NMS	Consumer Cyclical	Resorts & Casinos
498	NYQ	Consumer Cyclical	Auto Parts
499	NMS	Technology	Semiconductors
500	NYQ	Basic Materials	Agricultural Inputs
501	NYQ	Industrials	Specialty Business Services

	Currentprice	Marketcap	Ebitda	Revenuegrowth	Country \
0	254.49	3846819807232	1.346610e+11	0.061	United States
1	134.70	3298803056640	6.118400e+10	1.224	United States
2	436.60	3246068596736	1.365520e+11	0.160	United States
3	224.92	2365033807872	1.115830e+11	0.110	United States
4	191.41	2351625142272	1.234700e+11	0.151	United States
..	...	...	...	...	...
497	32.82	6973593600	3.668000e+09	-0.040	United States
498	31.88	6972155904	1.882000e+09	-0.048	United States
499	70.85	6697217024	6.731300e+08	-0.052	United States
500	50.15	6260525568	7.033000e+08	0.085	United States
501	19.17	4664099328	4.330000e+08	-0.031	United States

	Fulltimeemployees	State
0	164000.0	CA
1	29600.0	CA
2	228000.0	WA
3	1551000.0	WA
4	181269.0	CA
..	...	...
497	51000.0	NV
498	39900.0	MI
499	8700.0	NC
500	5800.0	PA
501	NaN	VA

[502 rows x 10 columns]

In [4]: `print(df.info())`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 502 entries, 0 to 501
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Exchange              502 non-null   object
1   Sector                502 non-null   object
2   Industry              502 non-null   object
3   Currentprice          502 non-null   float64
4   Marketcap             502 non-null   int64
5   Ebitda                473 non-null   float64
6   Revenuegrowth         499 non-null   float64
7   Country               502 non-null   object
8   Fulltimeemployees     493 non-null   float64
9   State                 482 non-null   object
dtypes: float64(4), int64(1), object(5)
memory usage: 39.3+ KB
None
```

```
In [5]: print(df.duplicated().sum())
```

0

```
In [6]: df = df.dropna()
```

```
In [7]: print(df.isnull().sum())
```

```
Exchange      0
Sector        0
Industry      0
Currentprice  0
Marketcap     0
Ebitda        0
Revenuegrowth 0
Country       0
Fulltimeemployees 0
State         0
dtype: int64
```

```
In [8]: print(df['Exchange'].unique())
```

```
['NMS' 'NYQ' 'BTS' 'NGM']
```

```
In [9]: print(df['Sector'].unique())
```

```
['Technology' 'Consumer Cyclical' 'Communication Services'  
 'Financial Services' 'Consumer Defensive' 'Healthcare' 'Energy'  
 'Industrials' 'Utilities' 'Real Estate' 'Basic Materials']
```

```
In [11]: print(df['Industry'].unique())
```

```
[ 'Consumer Electronics' 'Semiconductors' 'Software - Infrastructure'
  'Internet Retail' 'Internet Content & Information' 'Auto Manufacturers'
  'Insurance - Diversified' 'Discount Stores'
  'Drug Manufacturers - General' 'Credit Services' 'Oil & Gas Integrated'
  'Healthcare Plans' 'Household & Personal Products'
  'Home Improvement Retail' 'Entertainment' 'Software - Application'
  'Beverages - Non-Alcoholic' 'Telecom Services' 'Communication Equipment'
  'Restaurants' 'Information Technology Services' 'Diagnostics & Research'
  'Medical Devices' 'Tobacco' 'Medical Instruments & Supplies'
  'Aerospace & Defense' 'Farm & Heavy Construction Machinery'
  'Travel Services' 'Asset Management' 'Financial Data & Stock Exchanges'
  'Conglomerates' 'Utilities - Regulated Electric' 'Computer Hardware'
  'Insurance - Property & Casualty' 'Railroads' 'Apparel Retail'
  'Semiconductor Equipment & Materials' 'Oil & Gas E&P'
  'Footwear & Accessories' 'Integrated Freight & Logistics'
  'Insurance Brokers' 'Biotechnology' 'REIT - Industrial'
  'Utilities - Renewable' 'REIT - Specialty' 'Specialty Chemicals'
  'Electronic Components' 'Specialty Industrial Machinery'
  'Waste Management' 'Confectioners' 'Lodging'
  'REIT - Healthcare Facilities' 'Medical Care Facilities'
  'Specialty Business Services' 'Drug Manufacturers - Specialty & Generic'
  'Medical Distribution' 'Specialty Retail' 'Oil & Gas Midstream'
  'REIT - Retail' 'Building Products & Equipment' 'Insurance - Life'
  'Copper' 'Utilities - Diversified' 'Industrial Distribution'
  'Oil & Gas Equipment & Services' 'Engineering & Construction'
  'Utilities - Independent Power Producers' 'Rental & Leasing Services'
  'Oil & Gas Refining & Marketing' 'Grocery Stores' 'Gold'
  'Beverages - Brewers' 'Real Estate Services' 'Agricultural Inputs'
  'Airlines' 'Consulting Services' 'Electronic Gaming & Multimedia'
  'Trucking' 'Resorts & Casinos' 'Food Distribution'
  'Residential Construction' 'Packaged Foods' 'Health Information Services'
  'Building Materials' 'REIT - Residential' 'REIT - Diversified'
  'Scientific & Technical Instruments' 'Chemicals' 'Steel'
  'Pollution & Treatment Controls' 'Utilities - Regulated Water'
  'Farm Products' 'Electrical Equipment & Parts' 'Personal Services'
  'Utilities - Regulated Gas' 'Packaging & Containers' 'Solar'
  'Beverages - Wineries & Distilleries' 'Tools & Accessories'
  'Advertising Agencies' 'REIT - Office' 'Auto Parts' 'Luxury Goods'
  'Apparel Manufacturing' 'Auto & Truck Dealerships' 'REIT - Hotel & Motel'
  'Capital Markets' 'Pharmaceutical Retailers' 'Leisure'
  'Furnishings, Fixtures & Appliances']
```

```
In [8]: import numpy as np
df['Industry'] = np.where(df['Industry'].str.contains('Semiconductor Equipment & Materials|Software - Infrastructure
df['Industry'] = np.where(df['Industry'].str.contains('Drug Manufacturers - Specialty & Generic|Biotechnology|Medical
df['Industry'] = np.where(df['Industry'].str.contains('Insurance - Property & Casualty|Insurance - Life|Insurance Bro
df['Industry'] = np.where(df['Industry'].str.contains('Home Improvement Retail|Household & Personal Products|Restaura
df['Industry'] = np.where(df['Industry'].str.contains('Oil & Gas E&P|Oil & Gas Midstream|Oil & Gas Refining & Marketi

In [9]: print(df['Industry'].unique())

['Semiconductors' 'Consumer Goods' 'Business Sector' 'Medical field'
 'Industrial & utilities']

In [10]: print(df['Country'].unique())

['United States' 'Canada']

In [13]: print(df['Sector'].unique())

['Technology' 'Consumer Cyclical' 'Communication Services'
 'Financial Services' 'Consumer Defensive' 'Healthcare' 'Energy'
 'Industrials' 'Utilities' 'Real Estate' 'Basic Materials']

In [14]: import os
desktop = os.path.join(os.path.expanduser('~'), 'Desktop')
df.to_csv(r'C:\Users\KHAN\Desktop\sp-500(2).csv')
```